

Estimating I-Star Model Parameters

INTRODUCTION

In this chapter we introduce a framework enabling readers to build, test, and evaluate market impact models. The much celebrated “I-Star” model will be validated through this framework and techniques. Readers should experiment with alternative models and formulations of I-Star and determine the most suitable methodology given their trading and investing needs.

Our framework is based on the scientific method, a “process” quants use to achieve higher levels of knowledge. The scientific method works quite well—elementary school through Ph.D. dissertations. Yet, it appears, numerous Wall Street analysts seem to have abandoned the scientific in favor of models grounded in quark-gluon plasma ... —say another \$3 billion dollar hedging loss!

The scientific method is an “experimentation process” whereby analysts ask and respond to questions objectively. It provides the tools needed to uncover the truth through rigorous experimentation and statistical testing.

Off the record comment: Managers, if your analysts are not following the steps provided here it might be time to replace your analysts. Analysts, if your managers are not asking questions relating to this process, or they are not properly scrutinizing results, it might be time to find a new job. Industry professionals, if your vendors, brokers, consultants, or advisors are not providing essential background material, statistical evidence, and model transparency it is time to find new partners.

Our objective in this chapter is twofold: (1) teach model building and parameter estimation, and (2) help analysts expand their knowledge of market impact models—significantly.